

MATH 3551 A — Abstract Algebra I

Practice Midterm 1 — SOLUTIONS

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Fall 2026 • Total: 100 points

About this practice exam. It was drafted with the help of Claude Opus 5, running in a Cowork harness, working from previous exams in this course. **It has not yet been timed or proofread.** Like the real exam, it is built to take well under the time you are given: about thirty minutes of a fifty-minute midterm period, and about a hundred minutes of the final's two hours forty-five. The spare time is deliberate. *Please hunt for mistakes.* If you find a wrong answer, an ambiguous or unanswerable problem, a problem that leans on something we have not covered, or one that takes far longer than its point value suggests, tell me — in class, in office hours, or by email. I would much rather hear it from you now than print it on the real exam.

Problem 1. (30 points) **True or false; prove or disprove.**

- (a) **FALSE.** The order of a permutation is the lcm of its cycle lengths, so we would need a partition of at most 9 whose parts have lcm 18. Since $18 = 2 \cdot 3^2$, such a partition must contain a part divisible by 9 and a part divisible by 2; the smallest possibility is $9 + 2 = 11 > 9$. No such element exists.
- (b) **TRUE.** $|xy| = |x||y|$ for rational x, y , and the operation on $\mathbb{Q}^\times$ is multiplication, so $\varphi(xy) = \varphi(x)\varphi(y)$.
- (c) **TRUE.** $D_{16} = D_{2n}$ with $n = 8$. No reflection is central (if sr^i were central then $r(sr^i)r^{-1} = sr^{i-2} = sr^i$ forces $r^2 = 1$, false), and r^m is central iff $r^m s = sr^m$, i.e. $r^m = r^{-m}$, i.e. $8 \mid 2m$, i.e. $m \in \{0, 4\}$. So $Z(D_{16}) = \{1, r^4\}$.
- (d) **FALSE.** $(\mathbb{Z}/15\mathbb{Z})^\times$ has order $\varphi(15) = 8$, but every element satisfies $x^4 = 1$: indeed $\bar{2}^4 = \bar{16} = \bar{1}$, $\bar{4}^2 = \bar{1}$, $\bar{7}^4 = \bar{1}$, $\bar{11}^2 = \bar{1}$, $\bar{13}^4 = \bar{1}$, $\bar{14}^2 = \bar{1}$, $\bar{8}^4 = \bar{1}$. So there is no element of order 8.
- (e) **TRUE.** Choose the 3 moved points in $\binom{6}{3} = 20$ ways and arrange them in a cycle in $(3-1)! = 2$ ways: $20 \cdot 2 = 40$.
- (f) **FALSE.** $(\mathbb{Z}/8\mathbb{Z})^\times = \{\bar{1}, \bar{3}, \bar{5}, \bar{7}\}$ and each of $\bar{3}, \bar{5}, \bar{7}$ squares to $\bar{1}$, so every non-identity element has order 2; but $\mathbb{Z}/4\mathbb{Z}$ has an element of order 4. An isomorphism preserves orders of elements, so they are not isomorphic. (In fact $(\mathbb{Z}/8\mathbb{Z})^\times \cong V_4$.)

Grading note. Most of the credit here is in the *reason*, and the reason must be specific: a named counterexample, the actual permutation, or the actual count. A circled answer with no reason earns two of the five points; a wrong circle with a reason that is right about the underlying fact earns the three reason-points. Give full credit for a correct counterexample even if it is not the one printed above. The parenthetical remarks in each answer flag the error to expect on that item.

Problem 2. (25 points) **Inside $\mathbb{Z}/72\mathbb{Z}$.**

In $\mathbb{Z}/72\mathbb{Z}$ the order of $\bar{k}$ is $72/\gcd(k, 72)$, and $\langle \bar{k} \rangle = \langle \bar{d} \rangle$ where $d = \gcd(k, 72)$.

(a) $\gcd(30, 72) = 6$, so $|\overline{30}| = 72/6 = 12$.

(b) $\langle \overline{30} \rangle = \langle \bar{6} \rangle$, the unique subgroup of order 12:

$$\{\bar{0}, \bar{6}, \bar{12}, \bar{18}, \bar{24}, \bar{30}, \bar{36}, \bar{42}, \bar{48}, \bar{54}, \bar{60}, \bar{66}\}.$$

(c) $\varphi(72) = \varphi(2^3)\varphi(3^2) = 4 \cdot 6 = 24$.

(d) $\gcd(18, 72) = 18$, so $\langle \bar{18} \rangle$ is the subgroup of order $72/18 = 4$, namely $\{\bar{0}, \bar{18}, \bar{36}, \bar{54}\}$. Now $\langle \bar{k} \rangle = \langle \bar{18} \rangle$ if and only if $\bar{k}$ is one of the $\varphi(4) = 2$ generators of that cyclic group of order 4, i.e. $k \in \{18, 54\}$. (Equivalently: $\gcd(k, 72) = 18$.)

Grading note. Award the points for the *method*: $|\bar{k}| = n/\gcd(k, n)$ in (a), $\langle \bar{k} \rangle = \langle \overline{\gcd(k, n)} \rangle$ in (b) and (d), $\varphi(n)$ in (c). The expected error in (d) is to list all four elements of $\langle \bar{18} \rangle$ rather than only its two generators; that is worth 4 of the 7 points. A student who answers (d) by testing all 72 values gets full credit if the answer is right.

Problem 3. (25 points) **Two permutations.**

(a) $|\sigma| = \text{lcm}(3, 2, 4) = 12$.

(b) Apply τ first, then σ : $1 \mapsto 2 \mapsto 5$, $5 \mapsto 1 \mapsto 4$, $4 \mapsto 5 \mapsto 2$, $2 \mapsto 3 \mapsto 6$, $6 \mapsto 6 \mapsto 8$, $8 \mapsto 8 \mapsto 9$, $9 \mapsto 9 \mapsto 3$, $3 \mapsto 4 \mapsto 7$, $7 \mapsto 7 \mapsto 1$. Hence

$$\sigma\tau = (1\ 5\ 4\ 2\ 6\ 8\ 9\ 3\ 7),$$

a 9-cycle, so $|\sigma\tau| = 9$.

(c) The point 3 lies on the 4-cycle $(3\ 6\ 8\ 9)$, so σ^k fixes 3 if and only if $4 \mid k$. For $0 \leq k < 12$ this gives $k \in \{0, 4, 8\}$.

(d) $\sigma^{-1} = (1\ 7\ 4)(2\ 5)(3\ 9\ 8\ 6)$.

Grading note. (b) is the part that takes time and it is where the composition order matters; a student who computes $\tau\sigma$ instead gets $(1\ 5\ 3\ 8\ 9\ 6\ 2\ 4\ 7)$ or similar — award 3 of the 6 points for a correct computation in the wrong order, provided the convention used is stated. In (c) the intended observation is that only the cycle containing 3 matters; students who compute all twelve powers of σ get full credit but will not have time for problem 5.

Problem 4. (20 points) **Centralizer and normalizer in the quaternion group.**

- (a) $A = \langle i \rangle$, so $C_G(A) = C_G(i)$. Certainly ± 1 (central) and $\pm i$ (powers of i) commute with i . On the other hand $ji = -k$ while $ij = k$, so j does not commute with i ; the same computation applies to $-j, k, -k$. Hence

$$C_G(A) = \{1, -1, i, -i\} = A.$$

- (b) $ji j^{-1} = -i \in A$, and $j(-1)j^{-1} = -1 \in A$, so $jAj^{-1} = A$ and $j \in N_G(A)$. Since $A \leq N_G(A)$ as well and $|N_G(A)|$ divides 8 and is more than 4, we get $N_G(A) = Q_8 = G$. (Alternatively: $|G : A| = 2$, and every subgroup of index 2 is normal.)

- (c) $|G : A| = 8/4 = 2$, and the two left cosets are

$$A = \{1, -1, i, -i\}, \quad jA = \{j, -j, ji, -ji\} = \{j, -j, -k, k\}.$$

They are disjoint, each of size 4, and their union is G : $2 \cdot 4 = 8 = |G|$, as Lagrange requires.

Grading note. As in the real exam, the point is that $C_G(A) \neq N_G(A)$: the element j inverts i rather than fixing it. A student who reports $C_G(A) = G$ “because Q_8 is almost abelian” has missed the entire problem. Give full credit in (b) for the index-2 shortcut. In (c) accept $\{j, -j, k, -k\}$ in any order; note $ji = -k$, so a student who writes $jA = \{j, -j, k, -k\}$ without computing ji has still got the right set.